

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459898.67 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.018
Transit depth (Rp/Rs)^2	3.2 +/- 0.65 [%]
Orbital Inclination (inc)	84.29 +/- 1.86
Airmass coefficient 1 (a1)	0.987 +/- 0.013
Airmass coefficient 2 (a2)	0.011 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.32 %
Best Comparison Star	#2 - [208, 31]
Optimal Aperture	3.18
Optimal Annulus	10.43
Transit Duration (day)	0.065 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.018 vs 0.1646 (deviation 0.75σ)
Duration fit vs published	0.065 d vs 0.0758 d (deviation 0.53σ)
Mid-time O–C	2.7 min
Depth SNR	9.3
Residual scatter	1.32 %

Light curve

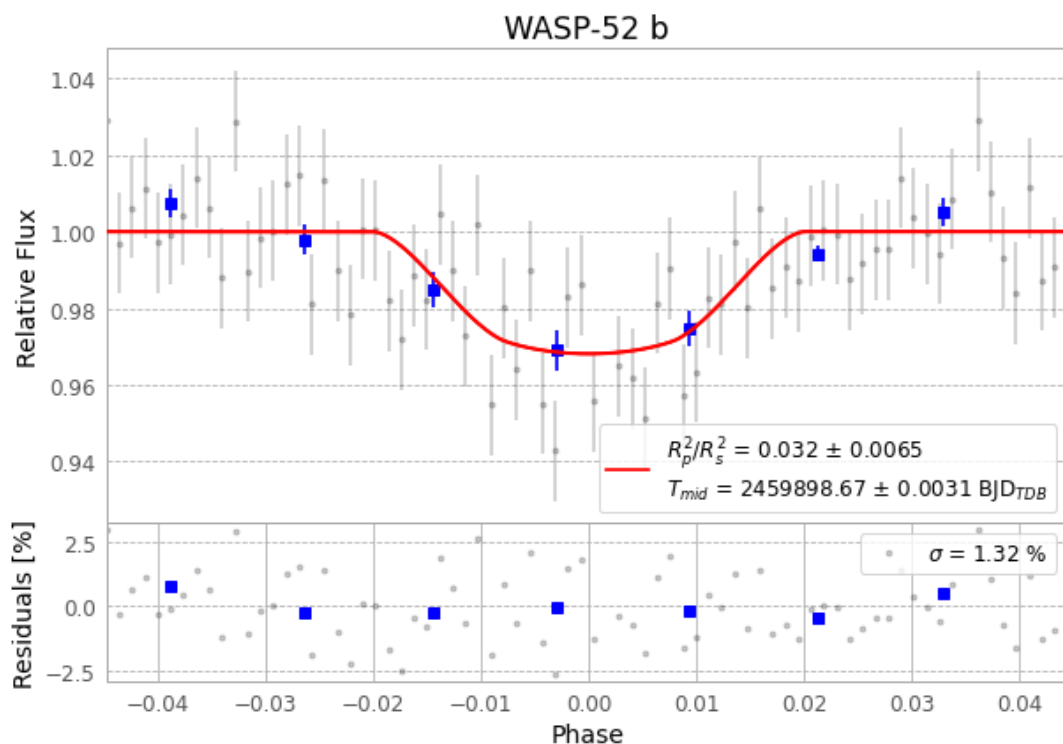


Figure 1 — FinalLightCurve_WASP-52 b_2022-11-14.png (SHA-256 5930196308c7c504...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 151], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 21d7b1c887a17c57...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

76 FITS frames (50 MB), WASP-52221115020618.FITS → WASP-52221115055115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221115.log (060d8baa8b78...), FinalLightCurve_WASP-52 b_2022-11-14.png (5930196308c7...), FinalLightCurve_WASP-52 b_2022-11-14.csv (71a948c53318...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 03:24Z.